EQ. 9 **Fully expanded formula** · the two levels in one expression

$$MP_{norm} = \{[(\Sigma_i\, w_i\, MI_i) \times (F \times P \times MDA^{\mu})] \times (PW^{w_1}\, IW^{w_2}\, SocW^{w_3})^{1/\Sigma w} \times [(IC^{a_1}\, TE^{a_2}\, BC^{a_3})^{1/\Sigma a}]^{\gamma} \times [\min\{1, (\beta_1\, CS_m + \beta_2\, CD_m + \beta_3\, RC_m)(1 + \delta \cdot CDip)\}]^{\alpha}\}^{1/(3+\gamma+\alpha)}$$

$X_k \to 0$ Collapse

If any component tends to zero, so does MP_{norm} (every exponent is positive).

$\partial MP_{norm}/\partial X_k$
Increasing and concave

Strictly increasing and strictly concave in each component for any α, γ > 0: decreasing marginal returns everywhere.

$\partial MP_{norm}/\partial \alpha$ Sign of CP − G

Monotone in α with the sign of CP − G; as α → 0 the index tends to G and as α → ∞ to CP.

$MP_{norm} \in (0, 1]$ Bounded

Between min{G, CP} and max{G, CP}; interpretable through the five-level scale.

Note: substituting Eqs. 4–8 into Eqs. 2–3 yields the complete model. All parameters are calibratable by state without modifying the architecture; the model has been verified numerically (300,000 random draws) but neither calibrated nor externally validated. Verification script: verification/verify_model.py.